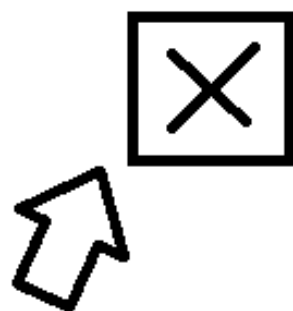
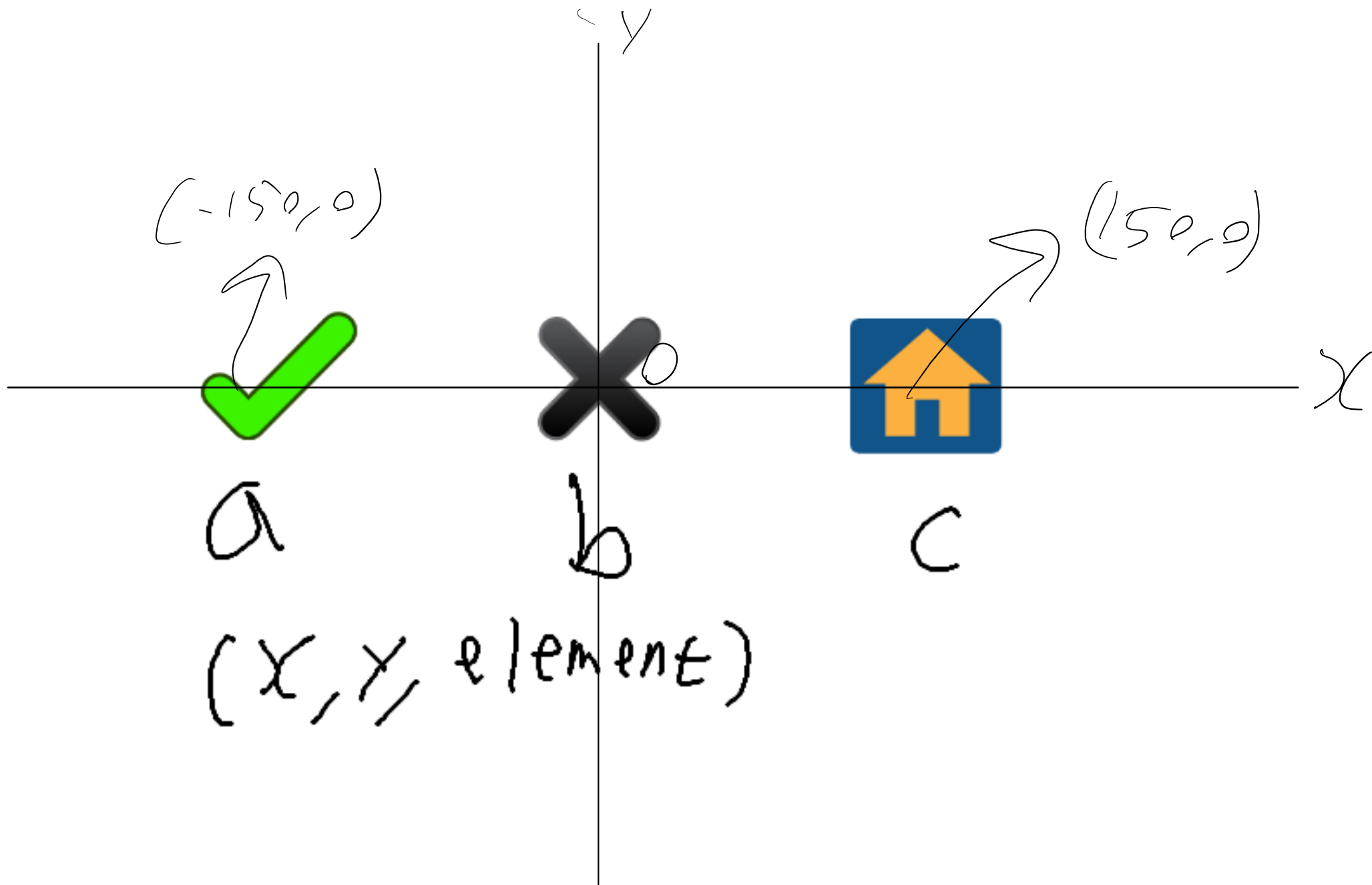


Improving User Interface Click System With AI

2024044620 김종진

AD





Getting data

classification

(xpos):(ypos):(element)

```
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import LabelEncoder

def load_data(file_path):
    X = []
    y = []
    with open(file_path, "r") as f:
        for line in f:
            line = line.strip()
            if not line:
                continue
            x, y_val, label = line.split(":")
            X.append([float(x), float(y_val)])
            y.append(label)
    return np.array(X), np.array(y)

def train_model(X, y, model_type="rf"):
    le = LabelEncoder()
    y_encoded = le.fit_transform(y)
    X_train, X_test, y_train, y_test = train_test_split(
        X, y_encoded, test_size=0.2, random_state=42
    )
    model = RandomForestClassifier(n_estimators=100, random_state=42)
    model.fit(X_train, y_train)
    accuracy = model.score(X_test, y_test)
    print("accuracy: " + str(accuracy))
    return model, le

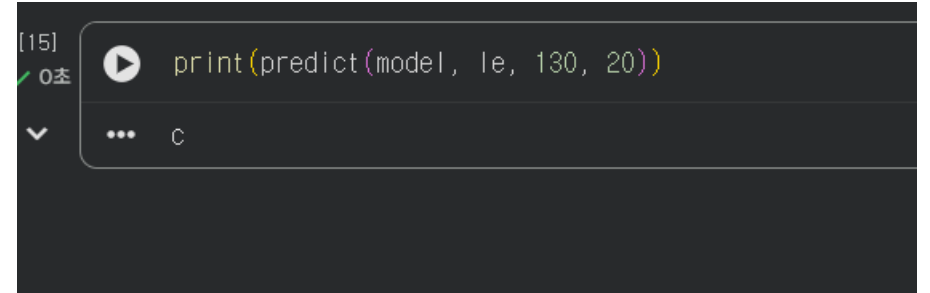
def predict(model, le, x, y):
    pred = model.predict([[x, y]])
    label = le.inverse_transform(pred)
    return label[0]

X, y = load_data("x,y,e.txt")
model, le = train_model(X, y, model_type="rf")
```

```
1:-14:b
-3:-14:b
-7:-11:b
-8:-9:b
-13:-6:b
-14:-3:b
-15:-1:b
-15:0:b
-13:3:b
117:29:c
118:29:c
133:29:c
147:29:c
151:25:c
162:25:c
169:20:c
173:16:c
174:5:c
174:-9:c
174:-19:c
166:-21:c
152:-19:c
133:-19:c
126:-18:c
125:-13:c
125:-5:c
```

Using Machine Learning

- Training Data
- Input : (xpos,ypos) when you click
- Output : (element)
- So AI can distinguish clicking ad itself or ad-dismiss button
- miss-click-free click



A screenshot of a Jupyter Notebook interface. It shows a code cell with the following Python code: `print(predict(model, 1e, 130, 20))`. The cell is numbered [15] and has a green checkmark and '0초' (0 seconds) indicating it has been executed successfully. Below the code cell, there is a dropdown menu with three dots and the letter 'C'.

<https://github.com/jongjinhy/aiinterfacetest/blob/main/aiinterfacetest.ipynb>